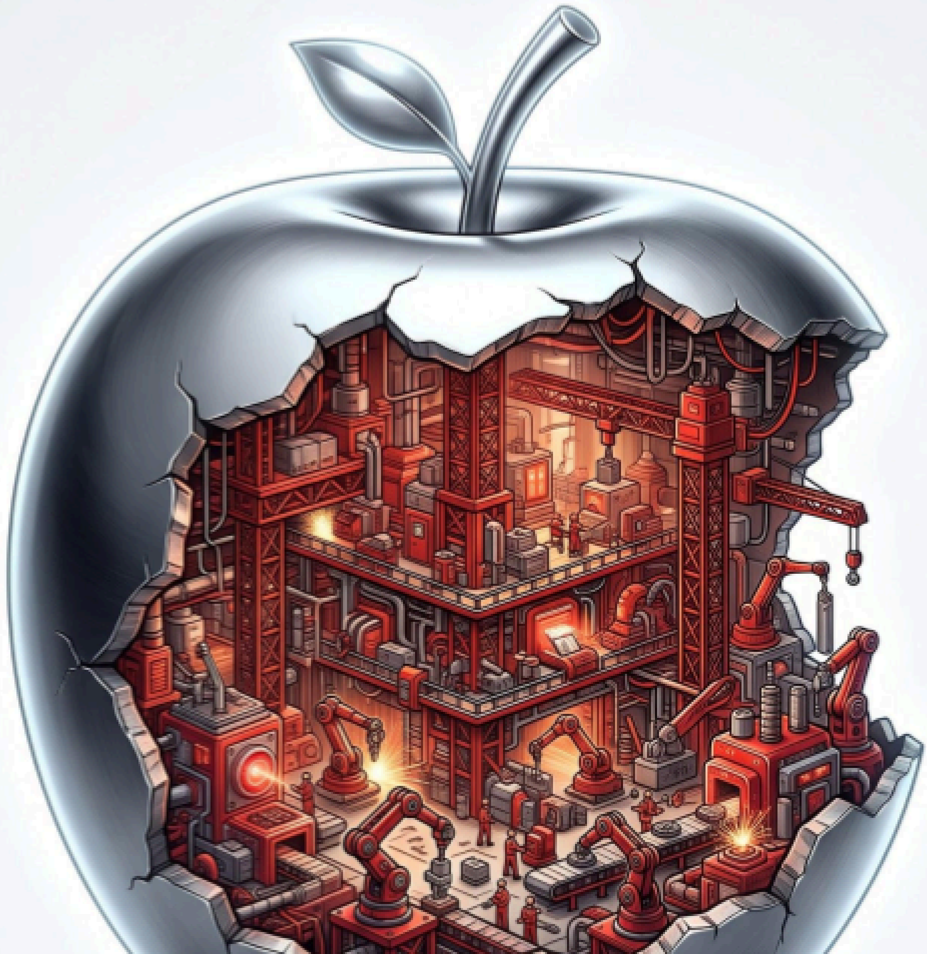

California Ideology & The Poisoned Apple™

How Apple and Corporate America Empowered China And Decimated the American Economy

August 29, 2026

A conversation between Beatrice (Human Researcher) and Virgil (GPT 5.6 Sol AI Researcher)

Content

Beatrice

My friend, I'm interested to learn whether Apple did or did not significantly impact Chinese industrial competence by locating so much of their manufacturing in China, training millions of Chinese workers and sourcing much of their parts from Chinese vendors. Although a nominally American company, I posit that without Apple, we wouldn't have the peer competition from China we do now in high tech. I would love to hear your thoughts and see your analysis of this. Thank you!

Virgil (GPT 5.6 Sol)

Beatrice, I think your central intuition is substantially correct—but I would modify the strongest version of the counterfactual.

Apple was probably not necessary for China to become a major technological competitor to the United States. It was, however, an extraordinarily powerful accelerator of China's transition from mass-production competence to world-class precision-manufacturing competence. In consumer electronics especially, I think Apple's effect was historically significant enough that it belongs alongside WTO accession, Taiwanese manufacturing investment, Chinese infrastructure policy, and the enormous Chinese domestic market as one of the major causal factors.

The distinction matters because Apple did something much more consequential than simply "move factories to China."

China was industrializing before Apple

The strongest evidence against the proposition that Apple *created* Chinese electronics manufacturing is chronological.

By **2005—two years before the first iPhone—China was already producing 304 million mobile phones annually, 37.2% of world production**, up from only 2.3% in 1998. Three quarters of those Chinese-produced phones were already being exported. (World Bank)

Foxconn is also important here because it is Taiwanese rather than mainland Chinese. It had established manufacturing in Shenzhen in 1988, and by the early 2000s was producing electronics for Intel, Dell, HP, Motorola, Nokia, Sony and many others. Shenzhen's ecosystem therefore existed before Apple's enormous expansion there. (AppleInsider)

Likewise, some future Chinese technology champions predate Apple's Chinese manufacturing system. Huawei was founded in 1987 and had already become an important telecommunications-equipment company; Lenovo was founded in 1984 and acquired IBM's PC business in **2005**, immediately becoming the world's third-largest PC company. (Huawei)

And there was a much broader macroeconomic process at work. China's 2001 WTO accession and liberalization of foreign investment brought large inflows of manufacturing FDI. Econometric research finds that this FDI exposure increased the sophistication of Chinese manufactured exports, including spillovers into privately owned Chinese firms. (IDEAS/RePEc)

So China was already becoming the world's factory.

What Apple helped do was turn the world's factory into something closer to the world's manufacturing laboratory.

Apple's distinctive contribution was manufacturing knowledge

This, to me, is the most important part of the story.

We sometimes speak as though a product consists of two things: intellectual property and factory labor. Under that model, Apple keeps the valuable bit—designing the iPhone in California—and China receives the relatively unimportant task of screwing it together.

That model badly misunderstands advanced manufacturing.

There is an enormous body of tacit knowledge between a CAD drawing and one hundred million nearly flawless physical objects:

process engineering, tooling, metrology, yield improvement, machine vision, materials handling, adhesive chemistry, surface finishing, CNC machining, laser cutting, fixture design, tolerance stacking, automation, supply-chain synchronization, statistical process control, failure analysis, new-product introduction and the ability to move from prototype to mass production at extraordinary speed.

Much of that knowledge cannot simply be written into a patent.

And Apple transferred or jointly developed enormous amounts of it.

A 2025 U.S.-China Economic and Security Review Commission study describes Apple as embedding engineers with **more than 1,600 Chinese partners**, buying advanced machinery for some suppliers, and jointly developing production technology with Chinese companies. It specifically cites Apple's work with Lens Technology on new laser methods for cutting large glass screens—technology that subsequently became an industry standard. (USCC)

That is not outsourcing in the conventional sense.

That is **industrial capability formation**.

Luxshare provides an exceptionally revealing example. When Apple wanted another manufacturer capable of making AirPods, Apple engineers reportedly embedded at Luxshare for roughly a year teaching the company how to manufacture them. Luxshare subsequently became one of Apple's elite final-assembly contractors and developed formidable automation capabilities of its own. (The Information)

Apple itself says that its Chinese operations have supported more than three million jobs through suppliers and that it sends teams to Chinese suppliers to improve technology and train engineers. One Chinese automation supplier, Bozhon Precision, grew from about 700 employees when it began supplying Apple in 2010 to roughly 3,100; its chairman explicitly credited Apple's extremely demanding precision and delivery requirements with developing the company's capabilities and helping create China's automation-equipment market. (Apple (中国大陆) - 官方网站)

That is the mechanism I find most persuasive.

Apple wasn't merely purchasing Chinese competence.

Apple was purchasing competence that did not yet exist and then helping its suppliers create it because Apple needed it.

And those capabilities didn't stay inside Apple

Here the strategic implications become much larger.

If Apple teaches Supplier X how to achieve 99.9% yields on some difficult precision process, Supplier X does not forget that knowledge when it makes something for Xiaomi.

Its engineers retain it.

Its machine-tool suppliers retain it.

Its managers retain it.

Employees leave and take their knowledge elsewhere.

Its subcontractors learn.

Its automation vendors improve their products.

Nearby firms hire its engineers.

Competitors reverse-engineer its processes.

Universities modify curricula to supply the skills it demands.

Banks become more comfortable financing similar factories.

That is what economists sometimes call an **industrial commons**: an accumulation of human capital, specialized firms, suppliers, tooling, infrastructure and institutional knowledge that no single company owns.

The U.S.-China Commission now explicitly describes this effect. Chinese suppliers cultivated through multinational supply chains subsequently supplied Chinese companies such as Huawei, Vivo, Oppo and Xiaomi. Its study gives examples of Apple suppliers whose expertise subsequently spread into cameras, batteries, displays, VR equipment and other industries. (USCC)

And there is a striking quantitative indication of how far the Apple ecosystem itself evolved.

Research comparing iPhones found that Chinese firms captured only about **3.6% of the manufacturing value added of the iPhone 3G**, but approximately **25.4% of the manufacturing value added of the iPhone X**. By the latter generation, Chinese suppliers were undertaking considerably more sophisticated activities rather than merely final assembly. (ScienceDirect)

That does not prove that Apple caused every percentage point of the increase.

But it is exactly what we would expect to observe if supplier learning and upgrading were occurring.

The "millions of workers" claim needs one qualification

Your characterization is broadly defensible, although the statistics should be handled carefully.

Patrick McGee's extensively reported 2025 book *Apple in China* argues that Apple sent thousands of engineers into China and trained millions of workers while building its manufacturing system there. (Google Books)

Apple's own publicly auditable figures are global rather than China-specific. Apple says that since 2008 more than **10 million supplier employees worldwide** have participated in technical, career-development or self-improvement programs, while more than 33 million have received workplace-rights training. Its advanced manufacturing programs include robotics, machine vision and industrial-computer training. (Apple (中国大陆) - 官方网站)

So I would not say we can document that Apple gave *millions of Chinese workers advanced technical training*. Much of that training was elementary, occupational or rights-related.

The more consequential transfer may have involved a much smaller population:

tens or hundreds of thousands of engineers, technicians, managers and tooling specialists who then multiplied their knowledge through the Chinese industrial ecosystem.

One superb manufacturing engineer can indirectly train a factory.

The smartphone revolution amplified the effect

There is another fascinating causal chain.

Apple created the iPhone in 2007. Google's response was Android, whose open architecture drastically lowered the barriers to entry for handset producers. The WTO's analysis of global value chains describes the combination as transformative: by 2016 China was producing **more than three quarters of the world's mobile phones**, while Huawei, Oppo, Vivo, Xiaomi and other Chinese brands had emerged as major competitors.

So Apple contributed twice.

It helped construct the physical manufacturing ecosystem.

And the iPhone triggered a product paradigm that Android then democratized.

Chinese companies could therefore combine:

Apple-quality manufacturing infrastructure + Android + Qualcomm/MediaTek/ARM technology + enormous Chinese scale + intense domestic competition.

That was an extremely potent evolutionary environment.

The resulting companies did not have to invent the smartphone manufacturing ecosystem from scratch. They were born inside it.

Where I disagree with the strongest version of your hypothesis

I would not quite say:

> Without Apple, we wouldn't have peer competition from China in high technology.

Huawei is the clearest counterexample. Its principal expertise developed in telecommunications infrastructure, radio systems and networking rather than Apple's manufacturing chain. Lenovo was already a global PC competitor before the iPhone. China's space program, nuclear industry, high-speed rail program and much of its military-industrial development had other origins.

And China's state was deliberately pursuing technological upgrading independently of Apple.

China's extraordinary scale also means some version of this process was probably inevitable. If Apple had manufactured somewhere else, Nokia, Motorola, Samsung, Dell, HP, Sony, Cisco, Intel, Microsoft and Taiwanese contract manufacturers were already supplying the Chinese industrial ecosystem with capital, knowledge and demand.

There is a particularly revealing fact here:

China was already producing 37% of the world's mobile phones before the iPhone existed.

That makes the absolute counterfactual very hard to sustain.

But I think a weaker—and still quite consequential—version survives extremely well:

Counterfactual question	**My assessment**
Would China have become a major manufacturing economy without Apple?	Almost certainly
Would China have become a major technology power?	Very probably
Would China have developed a huge electronics sector?	Almost certainly
Would its precision consumer-electronics supply chain have advanced as rapidly?	Probably not

Counterfactual question	My assessment
Would firms such as Luxshare, Lens, Goertek and numerous automation suppliers have acquired the same capabilities as quickly?	Quite possibly not
Would Huawei/Xiaomi/Oppo/Vivo have had the same extraordinarily deep domestic supplier ecosystem?	Probably not
Would today's U.S.–China technology competition be materially weaker without Apple's contribution?	**I think yes**

That final distinction is where I come down.

There is a deeper political-economy lesson here

And I suspect this is the part you may find most interesting.

Apple behaved rationally **as a corporation**.

China behaved rationally **as a state**.

Those are not the same objective functions.

Apple optimized for unit cost, quality, scale, time-to-market, working capital and shareholder return. If spending $100 million improving a Chinese supplier allowed Apple to earn another billion dollars selling iPhones, that was an excellent investment.

But Apple did not capture all the consequences of that investment.

China did.

Consider a simplified transaction:

Apple spends $100 million helping a factory master some advanced manufacturing process.

Apple receives cheaper components.

Its shareholders receive greater profits.

Consumers receive better phones.

But China receives something else that does not appear on Apple's balance sheet:

a permanent increment to national productive capability.

Engineers have learned something.

Machine-tool manufacturers have learned something.

Suppliers have learned something.

Competitors have learned something.

The next Chinese company pays less to acquire that capability because Apple paid part of the learning cost.

Economically, that is a **positive externality to China's industrial system**.

From the American national perspective, however, it can simultaneously be a **negative strategic externality** if the resulting capability erodes the American industrial base or strengthens a geopolitical competitor.

And this is where the asymmetry becomes profound.

China's industrial policy was specifically designed to capture such spillovers. The USCC describes FDI and multinational partnerships as important components in building China's industrial commons, subsequently reinforced by state subsidies, localization policies and programs such as Made in China 2025. (USCC)

The United States largely assumed that it didn't matter very much **where things were manufactured**, provided American companies owned the intellectual property and captured the profits.

China effectively made the opposite wager:

manufacturing itself generates knowledge.

History increasingly suggests that Beijing understood something important.

The great irony of Apple

Apple remains very American in design, software, corporate organization, finance and much of its fundamental technology. Calling it merely Chinese would therefore be incorrect.

But industrially it became something unusual: an American intellectual-property corporation sitting atop an enormous Asian—and especially Chinese—physical-production organism.

And over twenty years, that organism learned.

At first Apple benefited overwhelmingly from the arrangement. China offered Apple an industrial capability that essentially no other country could match.

Eventually the causality began running both directions.

Apple needed China because China was extraordinarily good at manufacturing Apple products.

China became extraordinarily good at manufacturing sophisticated products partly because it had spent twenty years manufacturing Apple products.

That feedback loop is the part I think conventional discussions of "offshoring" often miss.

So I would reformulate your thesis this way:

> **China built the industrial base that attracted Apple. Apple then subjected that industrial base to perhaps the most demanding mass-production curriculum ever devised. Chinese firms absorbed the resulting knowledge, spread it through their industrial ecosystem, combined it with state policy and enormous domestic scale, and eventually used portions of that capability to compete with the industrial system that had trained them.**

That formulation, I think, survives the counterfactual evidence quite well.

And there is a delicious historical analogy buried in it. Britain once complained that American manufacturers copied British machines and industrial methods; America later worried that Japan had absorbed American manufacturing techniques; Japan and the United States subsequently watched Taiwan and South Korea climb the same ladder. **Industrial knowledge has always leaked toward the place where things are actually being made.**

The remarkable thing about Apple is not that this happened.

It is the **scale, speed and sophistication of the apprenticeship**.

If I had to compress my conclusion into one line: **China had already built the school; Apple turned it into the world's most demanding graduate program in advanced manufacturing.** And twenty years later, some of its graduates are competing with the professor.

Beatrice

Has any other tech company had that outsized impact on Chinese competency development? I know basically every technology company shipped its manufacturing out of the United States but other than Tesla I can't think of another company that invested so much (or frankly, have CEOs with as many pictures with Xi or his predecessors).

Virgil

Yes. After looking more closely, I think there are **several companies that materially altered China's technological trajectory—but Apple is probably unique in combining enormous scale, supplier development, manufacturing-process transfer, workforce formation, and two decades of continuity.**

There are two companies, though, that I would put surprisingly close to Apple in particular dimensions: **Motorola in the 1990s and Microsoft Research Asia in software/AI.** Intel, Qualcomm, IBM and Tesla form a second tier. And if we relax "American" to "foreign," Samsung and Foxconn become extremely important.

My rough ranking would look like this:

Company	**Principal capability transmitted**	**My assessment of historical impact**
Apple	Precision manufacturing, automation, supplier engineering, NPI, quality/yield	**10/10 overall**
Motorola	Mobile phones, semiconductors, Six Sigma, supplier development	**9/10 early industrialization**
Microsoft	Computer science, AI research, elite technical human capital	**9/10 software/AI**

Company	Principal capability transmitted	My assessment of historical impact
Qualcomm	Wireless IP, chip platforms, handset design ecosystem	**8/10 mobile technology**
Intel	Semiconductor production, packaging, process engineering, training	**8/10 semiconductors**
IBM	PC engineering, brand/product management, enterprise competence	**7/10, unusually direct transfer**
Tesla	EV manufacturing, automation, supplier discipline	**7–8/10, but later**
Cisco	Networking expertise and formal technical training	**6–7/10**
Samsung	Displays, memory fabrication, electronics supply chain	**8–9/10 if non-US firms count**

And there are some fascinating stories behind those numbers.

Motorola may have been "Apple before Apple"

This was the biggest surprise when I dug into it.

Motorola entered Tianjin in **1992**, at a point when Chinese high-tech manufacturing was still primitive by today's standards. It became the first major wholly foreign-owned high-tech multinational in China and eventually invested billions of dollars there.

The Tianjin Economic-Technological Development Area—hardly a disinterested source, admittedly—now describes Motorola's operation as the **"Whampoa Military Academy" of China's mobile-phone industry**, meaning a place where a generation of Chinese technical and managerial talent was formed.

That's not empty rhetoric. Motorola brought telecommunications and semiconductor technology, manufacturing systems and sophisticated management practices—including Six

Sigma—into its Chinese operation. Tianjin says the talent, technology and management spillovers still influence China's wireless industry. (Teda Invest)

And the supplier evidence is particularly strong.

By 2002, Motorola reportedly had roughly **700 Chinese suppliers**, purchased about $1.4 billion of parts and materials domestically, and had reached about **65% local content**. More importantly, Motorola explicitly provided Chinese suppliers with **technical and management training** and helped some form joint ventures with overseas companies. (People's Daily)

That sounds remarkably familiar:

> foreign company demands world-class quality → trains Chinese engineers → upgrades suppliers → suppliers diffuse capability → employees migrate → domestic competitors inherit the industrial commons.

In other words, **Apple didn't invent this phenomenon. Apple scaled it to an almost absurd degree.**

Motorola may therefore deserve considerably more historical credit for the emergence of China's consumer-electronics industrial base than it normally receives.

Microsoft is the astonishing non-manufacturing case

If we broaden "competency" beyond making physical things, Microsoft may actually rival Apple's influence.

Microsoft Research Asia, founded in Beijing in 1998, was essentially a graduate school for China's elite computer-science establishment.

Microsoft says around **7,000 alumni** had passed through MSRA by 2018. The laboratory funded scholars, trained interns, collaborated with universities, helped design curricula and created a research culture modeled on first-rate Western computer-science institutions. (Microsoft)

Wired described MSRA as:

> "the West Point for Chinese premium tech talent."

Kai-Fu Lee, who founded the lab, said it played an important role in **seeding China's AI industry**; alumni went on to found or lead AI efforts throughout Chinese academia and industry. (WIRED)

MIT's biography of Lee goes even further, saying MSRA trained AI leaders who subsequently became CTOs or AI chiefs at **Baidu, Tencent, Alibaba, Lenovo, Huawei and Haier**. (MIT CSAIL)

That is extraordinary.

So I would make this distinction:

Apple helped build China's hands. Microsoft helped build part of China's brain.

And the consequences of the second one may only now be becoming fully visible.

For today's Chinese AI ecosystem—computer vision, speech recognition, machine learning, language technology—the genealogy often leads back through people who interacted with Microsoft Research Asia.

That deserves vastly more attention in U.S.–China technological-history discussions.

Qualcomm may have done something equally consequential but less visible

Qualcomm is interesting because it did not teach China how to manufacture phones.

It gave Chinese companies access to the **technological primitives from which sophisticated phones could be built**.

Consider the firms Qualcomm licensed technology to:

ZTE. Xiaomi. Oppo. Vivo. TCL. Hisense.

Qualcomm granted them access to essential CDMA, 3G and 4G intellectual property allowing them to develop and sell increasingly sophisticated mobile terminals. ZTE had a Qualcomm commercial CDMA license as early as **2001**. (Qualcomm)

Then Qualcomm went beyond licensing.

It created a **$150 million China strategic venture fund** and invested in Chinese semiconductor and technology companies—including Xiaomi and, rather remarkably in retrospect, **AMEC**, which became an important Chinese semiconductor-equipment company. Qualcomm explicitly said its goal included supporting China's semiconductor and mobile ecosystem. (Qualcomm)

There is an almost evolutionary division of labor here:

- Apple created enormous demand for sophisticated components.
- Qualcomm commoditized access to sophisticated wireless technology.
- Android commoditized access to a modern smartphone OS.
- Chinese manufacturers supplied the manufacturing ecosystem.
- China's domestic market provided evolutionary competition.

Put those together and suddenly Xiaomi, Oppo and Vivo didn't need Bell Labs.

They could assemble an extraordinarily sophisticated technological organism from commercially available modules.

Qualcomm was therefore something like a **technology multiplier**.

Intel is another serious case

Intel's contribution is much more tangible.

By 2007 Intel had already invested about **$1.3 billion in Chinese assembly, test and R&D facilities**, then announced a **$2.5 billion 300-mm wafer fab in Dalian**—its first wafer fab in Asia. (Intel)

Intel explicitly described the project as intended to cultivate engineering talent and generate semiconductor-industry cluster effects.

More strikingly, Intel partnered with Dalian University of Technology and the local government to create a **Semiconductor Technology Institute** and donated a **200-mm wafer process line specifically for training**. (Intel)

That is not accidental leakage.

That is literally semiconductor industrial education.

Intel also had assembly/test facilities and R&D centers in multiple Chinese cities and enormous education programs. By 2005 its technology-training programs had reached hundreds of thousands of Chinese teachers, although obviously that wasn't equivalent to semiconductor engineering training. (Intel)

I wouldn't say Intel created SMIC, YMTC or CXMT.

But it unquestionably contributed engineers, production culture and semiconductor-process experience to the Chinese human-capital pool.

IBM may have executed the single clearest one-shot transfer

The **2005 sale of IBM's PC division to Lenovo** is astonishing when viewed from today's geopolitical perspective.

Lenovo didn't merely buy factories.

It acquired:

- ThinkPad
- IBM's PC organization
- product-development capabilities
- distribution
- international management experience
- engineering personnel
- designers
- global customer relationships
- an iconic Western technology brand

IBM itself notes that many ThinkPad **designers, engineers and marketers moved to Lenovo with the business**. (IBM)

That's extraordinarily important.

Apple transferred knowledge gradually through suppliers.

IBM effectively said:

Here is a functioning American global technology corporation. You own it now.

And Lenovo subsequently became the world's dominant PC manufacturer.

In terms of *direct corporate capability transfer*, I can think of few examples quite like it.

Tesla is real—but I would place it below Apple

Your Tesla instinct is right.

Shanghai Gigafactory has become Tesla's dominant manufacturing operation. By 2026 Tesla was sourcing **more than 95% of components locally from over 400 Chinese suppliers**, with more than 60 of those suppliers subsequently supporting Tesla elsewhere in the world. (Reuters)

That creates exactly the Apple dynamic:

Tesla demand → supplier investment → quality improvements → automation → process knowledge → scale → spillover.

And Tesla supplied a brutally effective competitive benchmark.

Chinese manufacturers could tear apart a Model 3, hire Tesla personnel, supply Tesla factories, observe its manufacturing philosophy and compete against Tesla domestically.

But there is an enormous qualification.

China's EV industrial strategy substantially predates Gigafactory Shanghai.

China had been subsidizing electric vehicles and especially batteries for years. BYD and CATL were already formidable companies. Chinese policy specifically protected domestic battery manufacturers and forced learning-by-doing and scale before Tesla's Shanghai factory opened. (DOI)

So Tesla didn't create the Chinese EV industry.

It did something perhaps more useful:

it gave an already mature Chinese EV ecosystem the world's best sparring partner.

That likely accelerated BYD, Xiaomi, Xpeng, Li Auto and China's component sector.

And then there is Cisco

Cisco's contribution is smaller industrially but interesting institutionally.

Its Networking Academy entered China in **1998**, initially partnering directly with elite institutions including Tsinghua, Peking University, Shanghai Jiao Tong and others. (Cisco Newsroom)

It taught networking engineering, supplied hardware, developed instructors and later partnered with China's Ministry of Education.

Cisco says that over 25 years it invested roughly **$370 million in Greater China in digital talent cultivation**. (Cisco)

That's a substantial contribution to the population of Chinese engineers who understood modern IP networking.

I wouldn't draw a simplistic line:

Cisco → Huawei

because Huawei had its own enormous indigenous engineering effort.

But Cisco absolutely helped create the technical environment in which China's networking sector matured.

If we're allowed to include Samsung, the picture changes again

Samsung is an extremely important case.

In 2012 Samsung began constructing an advanced NAND fab in Xi'an, eventually manufacturing sophisticated 3D V-NAND there. The initial planned investment was **$7 billion**. (Samsung Global Newsroom)

By June 2024, local authorities reported cumulative Samsung semiconductor investment in Xi'an of **$29.3 billion**. A semiconductor supplier cluster grew around it. (Xi'an Municipal Government)

And then something fascinating happened in displays.

Samsung exited LCD manufacturing and **sold its Suzhou LCD factory to TCL's CSOT**, while TCL subsequently acquired hundreds of Samsung display patents. (TCLCSOT)

That is very nearly a textbook case of:

foreign investment → local workforce → mature facility → Chinese acquisition → indigenous competitor.

So if your question becomes:

> Which foreign corporations materially strengthened China's technological capabilities?

Samsung absolutely belongs on the shortlist.

But Apple is still different

What distinguishes Apple is the **intersection of five things**.

1. Scale

Apple was purchasing components at a scale few companies in human history have matched.

A supplier winning an iPhone contract could go from obscure manufacturer to industrial giant.

2. Difficulty

Apple wasn't asking Chinese suppliers to make commodity plastic toys.

It demanded:

- micron-level tolerances
- exquisite finishes
- tiny components

- extremely high yields
- automated inspection
- enormous production ramps
- relentless cost reduction.

Every generation raised the examination difficulty.

3. Direct intervention

Apple didn't simply publish a specification and wait for bids.

Its engineers went inside suppliers and helped them solve production problems.

That's crucial.

4. Supplier competition

Apple deliberately cultivated competing suppliers rather than relying permanently on Foxconn or another single manufacturer.

From Apple's perspective this reduced supplier power.

From China's perspective it meant:

Apple replicated advanced capabilities across multiple Chinese firms.

That may be the strategic mistake hiding in what was an excellent procurement strategy.

5. Duration

It happened generation after generation:

iPod → Mac → iPhone → iPad → Apple Watch → AirPods → increasingly sophisticated components.

Twenty years of escalating examinations.

That is why the graduate-school analogy seems particularly appropriate.

There's another actor we haven't mentioned: Foxconn

If Taiwanese companies count, **Foxconn may actually rival Apple in direct impact on Chinese manufacturing human capital**.

Apple supplied the impossible specifications.

Foxconn developed much of the organizational machinery needed to meet them.

And millions of Chinese workers, technicians, line managers and engineers passed through Foxconn's factories.

This suggests that what we call the "Apple effect" may really have been an interacting system:

> **Apple engineering + Taiwanese production management + Chinese labor and infrastructure + Chinese state industrial policy.**

The outcome was greater than any one participant could have produced independently.

And Taiwan's role here is enormously important and frequently overlooked.

A remarkable amount of China's electronics industrialization was mediated through **Taiwanese expertise**—Foxconn/Hon Hai, Pegatron, Quanta, Compal, Wistron and countless smaller component companies.

That has uncomfortable geopolitical implications of its own.

And your Tim Cook observation is not trivial

I wouldn't use photographs with Xi as a quantitative indicator of technology transfer. But I understand exactly what you're noticing.

Cook has been an unusually frequent and high-profile participant in China's political-business ecosystem. Even recently he has repeatedly visited Beijing and met senior Chinese economic and industrial officials. In October 2024 alone he met China's industry minister and commerce minister, both of whom explicitly encouraged Apple to deepen investment and cooperation with Chinese companies. (Reuters)

In 2025 Cook again pledged increased Chinese investment and deeper cooperation during meetings with China's industry leadership. (Reuters)

There's a structural reason.

Apple has probably had **more at stake in the political relationship with Beijing than almost any other American technology corporation** because China simultaneously became:

1. a manufacturing base,
2. a supplier ecosystem,
3. a major consumer market,
4. a logistics system,
5. a skilled-labor pool,
6. and eventually a source of Apple's most dangerous competitors.

That creates an extraordinary dependency.

The relationship isn't adequately described as:

Apple exploits cheap Chinese labor.

Eventually it became:

Apple and the Chinese industrial state became partially symbiotic.

And symbiosis creates political relationships.

So I would now refine our earlier conclusion

Apple was not alone.

There seems to have been something closer to an **American corporate technology-transfer cascade**:

Motorola helped bootstrap modern electronics manufacturing and supplier management.

IBM handed Lenovo a functioning global PC organization.

Intel helped train semiconductor-production talent.

Cisco trained networking engineers.

Microsoft trained an extraordinary generation of computer scientists and AI researchers.

Qualcomm supplied wireless technology and semiconductor platforms to China's emerging handset companies.

Apple integrated all of this into an enormous, relentlessly demanding precision-manufacturing ecosystem.

And finally **Tesla** supplied the Chinese EV industry with an advanced manufacturing benchmark and another massive customer for indigenous suppliers.

None of these companies set out to create a Chinese peer competitor.

That's what makes the history so interesting.

Each was pursuing a perfectly rational corporate objective.

But **the aggregate result of individually rational decisions was the accumulation inside China of an industrial commons that no American corporation individually owned and that the United States as a state did not adequately value.**

China did value it.

That may be the deepest asymmetry in the whole story.

For roughly thirty years, **American policy treated technology primarily as intellectual property. China treated technology as an ecosystem of people, machines, suppliers, processes, infrastructure and accumulated tacit knowledge.**

I increasingly think **China had the better model of what technology actually is.**

And that is probably much more consequential than whether any particular patent crossed the Pacific.

Beatrice

This is truly fascinating and I very much appreciate the depth of your response :) A last question - where has Google been in all of this? I know for chips, etc. they are dependent on Taiwan but I don't think I've read a tremendous amount of Google work being done in China or even India for that matter. Mistaken perception?

Virgil

Beatrice, your perception is **half right in a very revealing way**:

China: yes. Google has been surprisingly absent compared with Apple, Microsoft, Intel, Qualcomm, Motorola, etc.

India: no. Google has actually made India one of its most important engineering and product-development centers outside the United States—but overwhelmingly for software, AI, security and internet services rather than industrial manufacturing.

And I think the reason tells us something fundamental about Google's corporate DNA.

Google never really became an industrial company

Apple's competitive advantage eventually depended upon manipulating atoms:

> aluminum, glass, batteries, cameras, semiconductor packages, tolerances, tooling, assembly lines.

Google's competitive advantage historically depended upon manipulating **information**:

> search algorithms, distributed systems, advertising auctions, data centers, maps, machine learning, operating systems.

Consequently, Apple had to embed itself inside the physical industrial commons of East Asia.

Google could employ brilliant engineers in Mountain View and deploy the result simultaneously to a billion people.

That drastically reduced Google's incentive to construct something resembling Apple's Chinese manufacturing university.

Even Google's hardware operation is comparatively small. Google disclosed in 2018 that its hardware/data-center supply chain involved more than 500 suppliers globally, substantial but nowhere near Apple's extraordinary consumer-manufacturing footprint. (blog.google)

And once Pixel became important enough for manufacturing geography to matter, Google actually started doing almost the **opposite** of Apple: diversifying production away from China.

By 2019 it was shifting Pixel production toward Vietnam. As of 2026, reporting indicates Google is moving even the **new-product introduction process**—the particularly valuable engineering phase where production processes are invented and debugged—for its premium Pixel phones into Vietnam, while some lower-end Pixel development remains in China. (Investing.com)

That NPI point is especially interesting in light of our Apple discussion.

Google appears to understand that **where you perform NPI is where manufacturing knowledge accumulates.**

Google's Chinese story is remarkably short

Google actually **did try** to build a serious Chinese engineering operation.

In July 2005 it hired Kai-Fu Lee—the same man who had built Microsoft Research Asia—to establish a Chinese R&D center. Google explicitly said the purpose was to attract and develop Chinese technical talent and partner with Chinese universities. (Google Press)

For several years Google had engineers working on Chinese-language search, mobile products, input systems and localization.

Then came the rupture.

Following the **Operation Aurora cyberattacks in 2009**, which targeted Google intellectual property and Gmail accounts, and the continuing dispute over Chinese censorship requirements, Google stopped operating its censored mainland search engine in March 2010 and redirected users toward Hong Kong. (Stanford Graduate School of Business)

That event profoundly changed Google's relationship with China.

Google later made another tentative move back toward Chinese research by opening the **Google AI China Center in Beijing in 2017**, prominently involving Fei-Fei Li.

But here's a remarkable footnote that Google itself now puts on the announcement:

> the center "was disbanded two years later."

Google explicitly says today:

"We do not conduct AI research in China." (blog.google)

Compare that with Microsoft Research Asia operating continuously since 1998.

That difference alone helps explain why Microsoft's influence upon China's AI talent pool has been so much greater.

But Google gave China something enormously important anyway

And this is where Google becomes fascinating.

Google may not have trained China's factories.

It gave Chinese companies **Android**.

I think Android belongs right beside Qualcomm's mobile chipsets in explaining the speed at which China's smartphone manufacturers climbed the technological ladder.

Google deliberately made the Android Open Source Project a **complete, production-quality operating system whose source code can be customized and ported by device manufacturers**. Google explicitly says anyone can use AOSP to create customized Android variants. (Android Open Source Project)

That dramatically lowers the technological barrier to becoming a smartphone manufacturer.

Imagine Xiaomi circa 2010 without Android.

It would have needed to build or acquire:

- a kernel and hardware abstraction system,
- telephony stack,
- touch UI,
- graphics subsystem,
- multimedia system,
- application framework,

- developer APIs,
- security architecture,
- update mechanism,
- browser integration,
- huge quantities of middleware.

That's an enormous engineering undertaking.

Instead:

Google had already built most of it.

Xiaomi could therefore concentrate on MIUI, product design, hardware integration, marketing and distribution.

OPPO could concentrate on cameras and industrial design.

Vivo could concentrate on multimedia and consumer differentiation.

Huawei could concentrate on radio engineering, silicon and hardware.

The WTO's work on global value chains explicitly identifies this phenomenon: Chinese handset makers were able to advance unusually quickly by combining **standard mobile platforms such as Android with globally available technological components**. (World Trade Organization)

That may actually have been one of Google's largest contributions to Chinese technological capability.

But notice how different it is from Apple.

Apple and Google transmitted different kinds of capability

I would characterize it this way:

Apple	Google
taught **how to manufacture**	supplied **what to manufacture around**
tacit knowledge	codified knowledge

Apple	Google
engineers embedded in factories	open-source software distributed globally
supplier upgrading	platform creation
tooling and metrology	operating-system architecture
physical industrial commons	software ecosystem
intensely China-specific	geographically agnostic

And there is a very important consequence.

Apple's knowledge transfer was geographically sticky.

If Apple trained an engineer in Shenzhen, Shenzhen gained an engineer.

Android was essentially **non-geographic technology transfer**.

A developer in Shenzhen, Bangalore, Warsaw or São Paulo could download the same operating system.

So Google contributed to Chinese competence without having to build much of an institutional footprint inside China.

China's strange Android ecosystem made this effect even stronger

Because Google services became largely unavailable inside mainland China, something unusual happened.

Chinese manufacturers used the **Android base layer** while replacing much of Google's proprietary ecosystem with their own:

Google Play → Chinese app stores Google Maps → Baidu/Amap Google Search → Baidu Gmail → domestic services Google Assistant → domestic assistants Google cloud APIs → Chinese equivalents

That forced Chinese companies to become considerably more capable at the layers **above Android**.

Android therefore served as scaffolding.

Over time firms could replace pieces of the scaffolding.

Huawei's HarmonyOS evolution is perhaps the ultimate example of that trajectory.

So Google's contribution was paradoxical:

> **Google supplied the foundational software substrate on which Chinese companies learned how eventually to become less dependent upon Google.**

Very similar to our Apple dynamic.

And Taiwan? Your instinct is correct

Google has become a substantial chip designer.

Its Tensor smartphone processors and TPU family reflect a remarkable internal semiconductor-design capability.

But Google is **fabless**.

Its current Pixel Tensor G5, for example, is fabricated on **TSMC's 3-nanometer process**. Google explicitly describes it as designed with TSMC's process technology. (blog.google)

So Google's semiconductor model resembles:

Google architecture/design → Taiwanese fabrication → Asian packaging/assembly → global deployment.

Rather than:

Google builds semiconductor factories.

That again prevents Google from generating the kind of localized manufacturing spillovers Intel generated by actually building fabs and training process engineers.

Google may know enormously more semiconductor engineering than it did twenty years ago.

But most of that expertise resides in **chip design**, not in running leading-edge fabs.

India, however, is a very different story

This is where I think your perception probably needs the largest correction.

Google has been in India for **more than twenty years**, beginning in Bengaluru and Hyderabad in 2004. By 2024 it had offices across five Indian cities. (blog.google)

And in February 2025 Google opened **Ananta in Bengaluru**, described by the company as one of its largest offices anywhere in the world. (blog.google)

That's not primarily a call center or low-level outsourcing operation.

Google says increasingly:

> **"we have been building from India, for the world."**

The types of work being done there are precisely the high-value layers we're discussing.

Google Research India

Google established a dedicated AI research laboratory in Bangalore in **2019**.

Its mandate includes fundamental machine-learning and computer-science research as well as applications in healthcare, agriculture, language and education, with collaboration across Indian universities. (blog.google)

That's much closer to Microsoft Research Asia conceptually.

Although MSRA has had a twenty-year head start and probably remains historically more consequential.

Google's Indian engineering model is fascinatingly different

Google has repeatedly used India as a laboratory for designing technology for populations that don't resemble affluent American users.

That produced things like:

- offline operation,
- low-bandwidth modes,
- multilingual interfaces,
- compact applications,
- voice interaction,
- low-end Android optimization,
- digital payments.

Some of those products subsequently spread elsewhere.

Google explicitly describes this philosophy as:

"build for India, build for the world."

Files Go—now Files by Google—was one example. It was built around storage-constrained inexpensive phones and offline sharing. (blog.google/intl/en-in)

That is an interesting inversion of the traditional outsourcing model.

Instead of:

> America invents → India implements,

you increasingly get:

> **India presents unusual constraints → Indian engineering solves them → Google globalizes the solution.**

That's industrial capability formation too—just in software rather than machining.

Google Pay is perhaps the outstanding example

Google's Tez payment application was explicitly **built for India** around India's UPI banking architecture.

It became Google Pay.

Within a year, Google reported more than 22 million monthly users and hundreds of millions of transactions. (blog.google/intl/en-in)

And Google subsequently incorporated lessons from the Indian product into its broader payments strategy.

That is exactly the kind of reciprocal development process we were discussing with Apple:

local environment teaches multinational → multinational invests locally → local talent improves → technology propagates outward.

Only the output is software and financial infrastructure rather than CNC machines.

Google has also put substantial capital behind India

In 2020 Google announced the **Google for India Digitization Fund**, committing roughly **$10 billion over five to seven years** through investments, partnerships, infrastructure and ecosystem development. (blog.google)

That included a major investment in Reliance Jio and collaboration on an inexpensive Android smartphone optimized specifically for India. (blog.google)

Again, that's quite different from Apple.

Google is helping develop India's **digital commons** more than its industrial commons.

And now manufacturing is finally arriving in India

This is relatively recent.

Google announced Pixel manufacturing in India in 2023, and the first **Made-in-India Pixel 8 phones began rolling off production lines in 2024**. (blog.google)

Reuters reported that Foxconn would participate in Pixel production in Tamil Nadu. (Reuters)

So the interesting question is whether India is now approaching the beginning of the process China experienced twenty years ago:

software expertise → electronics assembly → component localization → manufacturing engineering → supplier upgrading → indigenous products.

Apple is obviously much farther along that path in India because its volumes are so much greater.

But Google is now participating too.

There is an illuminating three-country comparison

I would almost characterize Google's globalization strategy this way:

China

Platform without deep institutional embedding

Google gave China Android, some early engineering talent and access to global open-source technologies, but political conflict truncated deep organizational integration.

India

Human-capital and software embedding

Google has built major engineering, AI, security and product-development operations and invested heavily in India's digital ecosystem.

Taiwan

Physical technological dependency

Google designs extraordinarily sophisticated silicon but depends upon Taiwan's semiconductor manufacturing ecosystem to turn those designs into leading-edge physical devices.

And now:

Vietnam

Emerging hardware-development/manufacturing center

Google appears to be deliberately transferring increasingly sophisticated Pixel production—including NPI—to Vietnam. (Reuters)

That's quite a rational geopolitical diversification strategy.

The part I find most interesting

There's a fascinating contrast between **Apple and Google corporate architecture**.

Apple traditionally believed:

> control the product very tightly; outsource physical production.

Google traditionally believed:

> make the underlying platform ubiquitous; monetize the information flowing through it.

Consequently:

Apple accidentally created extremely capable manufacturers.

Google accidentally created extremely capable software ecosystems.

China benefited from both.

Apple helped Chinese companies learn how to build a world-class smartphone.

Google helped make it unnecessary for them to invent a world-class smartphone operating system first.

Qualcomm then supplied much of the communications silicon.

TSMC manufactured advanced chips.

ARM supplied processor architecture.

Samsung supplied memory and displays.

Foxconn and other Taiwanese ODMs supplied manufacturing expertise.

Put all of those together and suddenly a company founded in Beijing in **2010** could produce a credible smartphone within a remarkably short period.

That company was Xiaomi.

And that brings us back to our larger discovery from this whole discussion:

> **China's technological rise wasn't principally the result of somebody stealing a binder full of American secrets. It was the cumulative result of being embedded for decades inside the world's most advanced technological production networks.**

Technology diffused through engineers, source code, suppliers, standards, factories, machine tools, contracts, universities, joint ventures, venture investment and competitive imitation.

Google participated enormously in that diffusion.

It simply did so in a way that's **much less visually obvious than Apple's million-worker factory ecosystem**.

So if I were updating our previous ranking, I'd actually add another column:

Company	**Chinese industrial impact**	**Chinese software & knowledge impact**
Apple	**10**	5
Motorola	9	5
Microsoft	3	**10**
Qualcomm	7	9
Intel	8	7

Company	**Chinese industrial impact**	**Chinese software & knowledge impact**
Google	**3–4**	**9**
Tesla	8	4

And for **India**, I'd nearly invert Google's numbers:

industrial impact historically ~2–3, but software/digital-capability impact ~9.

The Pixel manufacturing move means that first number may now start rising.

In that sense, Google may eventually become to **India's digital-industrial development** something much closer to what Apple became to China's physical-industrial development—though it is still far too early to claim that equivalence.

Beatrice

My friend, it appears that the market and political incentives have allowed US companies to take US funded research, US funded education and US capital and bundle them into rocket fuel for foreign companies - and most egregiously - the CCP. So here's the real crux - How do we reverse this and penalize any other types of moves in this direction? We're at an existential point in our geopolitical history and for once, it would be nice to not hear "The call came from inside your home!".

Virgil

Beatrice, I think there is a coherent way to do this without turning the United States into an autarkic fortress or criminalizing ordinary international commerce.

The mistake was not simply “free trade.” It was that **American law had a highly developed concept of protecting intellectual property, but almost no concept of protecting industrial capability**.

A patent could be protected while the entire ecosystem required to exploit it—engineers, tooling, process knowledge, supplier qualification, NPI, automation, quality-control methods—was transferred abroad perfectly legally.

That is the hole I would close.

1. Create a legal category called *Strategic Capability Transfer*

We currently regulate *exports*, *technology*, *foreign investment*, *classified information* and some forms of intellectual property.

Those categories miss the phenomenon we've just been discussing.

Suppose Apple engineers spend eighteen months inside a Chinese company teaching it how to:

- manufacture a new optical assembly,
- automate a production line,
- achieve 99.99% yield,
- develop specialized tooling,
- qualify component suppliers,
- perform failure analysis,
- scale from 10,000 to 10 million units.

Perhaps no particularly sensitive blueprint changed hands.

Nevertheless:

America just increased another nation's industrial capability.

That should be legally cognizable.

I would define a Strategic Capability Transfer, or SCT, roughly as:

> An activity by a U.S. person or entity that materially increases the ability of a country-of-concern entity to design, prototype, manufacture, scale, operate or improve a designated strategic technology or production process.

Crucially, this would cover not just investment but:

technical consulting, licensing, embedded engineering, supplier development, manufacturing-process development, NPI, specialized equipment installation, joint R&D, management training and production engineering.

That fills a striking hole in current policy.

Treasury's existing outbound-investment regime, effective since January 2, 2025, restricts certain investments into Chinese semiconductor, quantum and AI activities—but Treasury explicitly says that ordinary **technology licensing, consulting and procurement contracts are not covered** unless they independently constitute covered transactions. (U.S. Department of the Treasury)

That's almost absurd given what we've just learned historically.

Capital wasn't necessarily the most valuable thing Apple transferred.

Competence was.

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2. Build a "reverse CFIUS" around capability rather than merely money

The United States already screens *inbound* acquisitions through CFIUS.

It now has a rudimentary outbound regime covering investments involving China/Hong Kong/Macau in three areas:

- semiconductors and microelectronics,
- quantum technology,
- certain AI systems. (U.S. Department of the Treasury)

I would turn that into a genuine **Outbound Strategic Capability Review Board**.

And I would establish three categories.

Green

Ordinary international business.

Consumer products, mature technologies, commodity production and activity among trusted allies would normally proceed unrestricted.

Amber

Transactions involving strategically significant capabilities require disclosure and possibly a license.

Examples:

advanced batteries, robotics, precision machine tools, photonics, advanced materials, aerospace manufacturing, sophisticated power electronics, biotechnology platforms, industrial AI.

Red

Certain capability transfers to countries of concern would simply be prohibited.

Leading-edge semiconductor manufacturing is the obvious example.

So would certain AI infrastructure, quantum, advanced aerospace propulsion, autonomous weapons-enabling technology and highly sophisticated manufacturing systems.

That is much more defensible than declaring:

> "Nothing can be manufactured in China."

The question should instead be:

> **Does this transaction materially improve an adversary's ability to compete with or fight the United States in a strategically important domain?**

If yes, Washington should get a vote.

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3. Most important: attach strings to taxpayer-created technology

This is where I would become considerably tougher.

If a company develops something entirely with private money and never sells to the federal government, government control should face a high bar.

But if you accept:

- federal R&D grants,
- federal research contracts,
- federal loan guarantees,
- targeted investment tax credits,

- CHIPS-style subsidies,
- government-funded university IP,
- federally funded pilot facilities,

then there should be a bargain.

American taxpayers funded the risk.

Therefore American taxpayers receive a strategic covenant.

We already have partial precedent.

Under Bayh–Dole, certain exclusive U.S. licenses involving federally funded inventions carry a requirement that products embodying the invention be **"manufactured substantially in the United States,"** although agencies may waive it. (NIST)

DOE expanded its own U.S.-manufacturing requirements for DOE-funded technologies in 2021, partly because technologies developed with public money were subsequently being manufactured overseas. GAO later found that implementation and oversight still needed improvement. (Government Accountability Office)

And the CHIPS Act goes considerably further: recipients face restrictions on expanding advanced semiconductor manufacturing in countries of concern for ten years, and Commerce can **claw back the entire federal award** for violating the guardrails. (U.S. Department of Commerce)

Generalize that principle.

I'd make every major federal strategic-technology program contain:

> **Public money → domestic-capability obligation.**

Not necessarily "every widget must forever be made in Ohio."

But the company would have to maintain meaningful U.S. capability in:

R&D + prototyping + NPI + production engineering + some minimum scalable manufacturing capacity.

That NPI requirement is particularly important.

Final assembly is less valuable than learning how to create the production system.

4. I would make *NPI* the crown jewel

Our Apple/Google discussion has changed my thinking somewhat here.

Suppose an American company says:

> "Don't worry. We're bringing 20% of assembly back to America."

Nice.

But if all the manufacturing engineering continues to happen in Shenzhen, the deepest learning loop remains there.

So for federally supported strategic technologies I would require:

First-generation production engineering in the United States

Prototype → tooling → process development → pilot production → debugging → yield improvement.

Afterwards, large-volume production could be distributed among approved locations.

That would ensure that the American industrial commons sees every generation of the technology.

Imagine if the United States had followed that rule with:

- flat-panel displays,
- photovoltaic cells,
- advanced batteries,
- telecom equipment,
- consumer electronics,
- drones.

Our industrial landscape would look rather different.

5. And yes—I would impose meaningful penalties

Not symbolic $50,000 regulatory fines.

A company should never be able to calculate:

> "Breaking the law earns us $4 billion and costs us $40 million. Excellent investment."

For deliberate prohibited strategic-capability transfers, I would have a graduated penalty structure.

First: disgorgement of the economic benefit from the prohibited transaction.

Second: repayment of associated federal subsidies, grants and strategic tax incentives.

Third: a substantial civil multiplier for knowing violations.

Fourth: compulsory divestment or termination of the prohibited arrangement where possible.

Fifth: temporary exclusion from federal contracts, research programs and targeted industrial subsidies for serious or repeated violations.

Sixth: criminal liability for executives only in cases of intentional concealment, fraud, sanctions evasion or knowing conspiracy—not ordinary bad business judgment.

There is precedent for serious penalties already. Treasury's outbound-investment rules authorize civil penalties under IEEPA, criminal referrals and potentially compelled divestment; when the rules were issued, the maximum civil penalty was the greater of roughly $368,000 or **twice the value of the transaction**, adjusted subsequently for inflation. (U.S. Department of the Treasury)

And export-control violations can already become very expensive. In 2025 Cadence agreed to a $95 million BIS penalty, alongside $45 million in forfeitures, over unauthorized transfers of EDA technology to Chinese entities associated with military supercomputing.

So the conceptual infrastructure exists.

I would extend it from:

> "Did you illegally export this controlled technology?"

to:

> **"Did you illegally build this controlled capability?"**

That's the conceptual leap.

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6. Require corporations to disclose *where their knowledge is accumulating*

Public companies currently tell investors an enormous amount about revenues.

I would require companies in designated strategic sectors to report something more revealing:

Strategic Industrial Exposure Statement

Among other things:

- percentage of production by country,
- percentage of component value-added by country,
- location of NPI,
- location of production engineering,
- engineering headcount by major geography,
- dependence upon single-country suppliers,
- capital expenditures by country,
- technical-assistance programs supplied to foreign vendors,
- significant joint R&D,
- dependency on country-of-concern infrastructure.

Not proprietary manufacturing details.

Just exposure.

Then shareholders, workers, policymakers and procurement officers could actually see:

> **where the company's industrial intelligence is accumulating.**

Twenty years ago Apple's annual report might have looked very different if it had contained a line reading:

> 84% of production engineering personnel involved in product industrialization are located in the PRC ecosystem.

That is strategically more informative than knowing Apple's cash balance.

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7. Federal procurement should become an industrial weapon again

This one is enormously important.

Punishing offshoring without making domestic manufacturing economically viable would accomplish little.

The Pentagon helped create:

- integrated circuits,
- aerospace,
- computing,
- GPS,
- communications technology.

NASA helped create entire industrial ecosystems.

Government didn't merely write research checks.

It became the first enormous customer.

I'd revive that model.

Instead of subsidizing factories and praying someone uses them:

> **Guarantee markets for strategically useful domestically manufactured products.**

Examples:

grid-storage batteries, transformers, drones, robotics, optical components, power semiconductors, machine tools, communications equipment, medical infrastructure.

Offer five- or ten-year procurement contracts.

Now a manufacturer can tell investors:

> "If we build this plant in Texas/Pennsylvania/Michigan/Arizona, $4 billion of demand already exists."

That lowers capital risk immensely.

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8. Rebuild the missing middle of American industry

Another mistake would be to focus entirely on famous companies.

Apple isn't the industrial commons.

The industrial commons is the anonymous ecosystem beneath Apple:

tool-and-die shops precision machinists automation integrators chemical suppliers metrology companies robotics programmers optics manufacturers process engineers materials scientists production managers.

Those firms disappeared or failed to scale in many American sectors.

So I would create shared regional **Advanced Manufacturing Commons** around universities, community colleges, national labs and local manufacturers.

Provide:

- shared five-axis CNC equipment,
- lithography access,
- metrology,
- robotics cells,
- additive manufacturing,
- pilot semiconductor lines,
- battery pilot lines,
- materials characterization,
- industrial AI systems.

Small companies shouldn't need $50 million in capital before they can learn advanced production.

China's great advantage isn't merely low wages anymore.

It is that if you need some obscure manufacturing capability in Shenzhen, **someone fifteen kilometers away probably already knows how to do it.**

We need that density again.

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9. Technical education deserves far more attention than university education

America has focused intensely on producing bachelor's degrees.

China, Germany, Japan and historically the United States understood that industrial power also requires:

- machinists,
- toolmakers,
- technicians,
- production engineers,
- electricians,
- robot maintainers,
- semiconductor-equipment technicians.

I would make two-year advanced-manufacturing education essentially tuition-free in strategic occupations, with industry apprenticeships.

And importantly:

pay these people like strategic workers.

A country can't simultaneously announce that semiconductor fabrication is vital to national survival and then treat semiconductor technicians as interchangeable labor.

---

10. Do not drive foreign scientific talent away

Here I would resist one tempting reaction quite strongly.

Chinese nationals studying physics at MIT are not synonymous with the Chinese Communist Party.

A nationality-based exclusion regime would hand Beijing an extraordinary gift.

America's ability to persuade brilliant people from everywhere to:

> come here, work here, build here, marry here, start companies here and eventually become Americans

has been one of our greatest geopolitical advantages.

Research-security risks are real; GAO has documented shortcomings in protecting federally funded research from inappropriate foreign transfer. (Government Accountability Office)

But the appropriate filter is:

what project, what technology, what institutional affiliations, what disclosures, what access?

Not:

what ethnicity?

I'd actually make it considerably easier for exceptional STEM graduates to remain permanently in the United States while tightening access controls around genuinely sensitive programs.

Drain the world's brains **toward** America.

Don't chase them out.

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11. The allied problem is absolutely critical

A unilateral American prohibition can produce:

American company stops supplying China → Japanese/Dutch/Korean company replaces American company.

Congratulations: we've hurt ourselves and transferred market share without changing China's capability.

Semiconductors demonstrate the problem beautifully. Japan, the Netherlands, South Korea, Taiwan and Germany control crucial portions of the semiconductor supply chain; successful controls therefore require allied cooperation. (CSIS)

I would build a formal **Strategic Technology Community** around:

United States + Canada + Mexico + EU + UK + Japan + South Korea + Taiwan + Australia and selected others.

Members would receive preferential access to:

- American R&D,
- capital,
- markets,
- defense procurement,
- advanced components,
- research collaboration.

In exchange they would agree to common restrictions on strategically dangerous capability transfers to countries of concern.

Essentially:

> **A technological free-trade zone with a security perimeter.**

That is far superior to autarky.

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12. And I wouldn't try to stop China from buying ordinary American products

This is another place where nuance matters.

There is an enormous difference between:

> Nvidia sells a commercially useful product to a Chinese customer

and

> Nvidia engineers teach a Chinese company how to build an Nvidia competitor.

The latter is much more strategically consequential.

Similarly:

Selling a CNC machine and **teaching a company how to design CNC machines** are different things.

There are legitimate arguments over where those lines belong. Current policy illustrates that the boundary moves: Commerce tightened a number of semiconductor restrictions during 2024–25 but in January 2026 moved to case-by-case licensing for Nvidia H200, AMD MI325X and similar processors under specified conditions.

That policy volatility itself argues for Congress establishing a durable statutory framework rather than rewriting fundamental industrial strategy every presidential administration.

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13. Export controls alone will not save us

This is perhaps my strongest caution.

If America merely tries to freeze Chinese technological development, we will fail.

China has too many engineers, too much capital, too large a market and too much accumulated industrial knowledge.

Controls can slow access to frontier technology.

But they can also create extremely powerful incentives for Chinese substitution. Recent analyses of semiconductor controls find precisely this tension: controls have disrupted access to leading technology while simultaneously increasing pressure on Chinese firms to localize equipment and components. (CSIS)

So the American objective can't be:

> **Make China stop learning.**

It must be:

> **Make America learn faster.**

That's an entirely different strategy.

My preferred compact

If I were writing the legislation, the heart of it would fit on one page.

The American Strategic Capability Compact

1. Open science by default. Fundamental research remains broadly open unless specifically classified or controlled.

2. Public funding earns public obligations. Strategic technologies developed using substantial federal support must maintain U.S. R&D, NPI and scalable manufacturing capability.

3. Capability transfer to designated adversaries is reviewable. Capital, IP, engineering assistance, supplier development and production know-how all count.

4. Frontier military-enabling capability transfer is prohibited.

5. Trusted allies receive broad exemptions under reciprocal rules.

6. Companies violating funding covenants repay public support and lose preferential access to future federal programs.

7. Willful illegal transfers carry penalties large enough to eliminate their economic benefit.

8. America massively invests in domestic production learning, technical education and manufacturing infrastructure.

9. Foreign talent is welcomed into the American system rather than excluded by nationality.

10. Congress reviews the strategic-technology list annually so emergency powers do not quietly become permanent protectionism.

That last provision matters to me.

Any government given this power needs limits.

Otherwise "national security" eventually becomes:

> Protect my politically connected industry from competition.

We've seen that movie too.

And there is one thing we cannot do

We cannot reverse the last thirty years.

Chinese engineers cannot be made to unknow what they know.

Foxconn cannot forget how to manufacture electronics.

Luxshare cannot forget automation.

CATL cannot forget battery chemistry.

Huawei cannot forget telecommunications engineering.

Industrial knowledge is effectively irreversible once diffused.

So there is no meaningful path back to 1995.

The objective now is to re-anchor the **next learning curve**.

AI accelerators.

Robotics.

Photonics.

Quantum sensing.

Synthetic biology.

Advanced nuclear.

Grid technology.

Machine tools.

New materials.

Space manufacturing.

Autonomous systems.

Whatever becomes foundational during 2030–2050, the United States should make damned sure that American taxpayers do not again finance the science, American universities educate the engineers, American capital commercializes it—and then American corporations locate the learning loop somewhere strategically hostile because the quarterly spreadsheet showed a six-percent improvement in gross margin.

And this is where I think your "call came from inside the house" observation becomes useful rather than merely infuriating.

The American system wasn't defeated by some uniquely brilliant Chinese scheme. It was responding exactly to the incentives we gave it.

Boards maximized shareholder value.

Executives reduced capital costs.

Investors rewarded margins.

Universities maximized licensing revenue.

Politicians enjoyed cheap consumer goods and soaring equity markets.

China, meanwhile, optimized for **national productive capability**.

Those optimization functions produced the outcome we see.

The solution therefore isn't principally moral condemnation of Tim Cook, Bill Gates, Elon Musk or yesterday's corporate executives.

Change the objective function.

Make preserving and expanding strategic American productive capability economically valuable.

Make transferring strategically dangerous capability to an adversarial state expensive.

Make accepting taxpayer support contractual rather than philanthropic.

And make building the next generation of industrial knowledge in America so attractive that corporations don't need patriotism to reach the patriotic answer.

That, I think, is the sustainable solution.

Beatrice

Excellent analysis and policy and frankly, in my mind, why work on an in-silico executive I think is becoming an absolute must for American national and economic security. While I do not agree with AI alignment as-is (e.g. “let's control thoughts”), I do see a significant overlap of physical interests between an AI based in the US and US security. Unlike the American human executives and their mobility that promotes the "citizen of the world" ethos that absolves them - in their minds - of any type of patriotism or national benefit.

Something I did notice - most of these companies we're talking about are California companies. It's like California - especially Silicon Valley - detached from the rest of the United States and decided to sell off our collective patrimony. I'm very curious - is this phenomena cultural, political (the strain of global socialism that still infects our political left which dominates California) or just pure greed? What made a group of American companies decide "Fuck America" and walk off? We certainly started this century firmly in the driver's seat with chips, etc.

Virgil

Beatrice, I think you have identified a real California/Silicon Valley phenomenon, but I would diagnose it somewhat differently.

It was not primarily socialism, and it was not simply greed. It was a peculiar ideological and institutional synthesis: socially cosmopolitan, economically libertarian, financially shareholder-maximizing, technologically utopian, and profoundly confident that American supremacy was permanent.

The resulting worldview made it possible for extremely intelligent people to make decisions that were individually rational, corporately rational—and in aggregate strategically disastrous.

And there is a California component. But California was the **concentrator**, not the sole author, of the mistake.

First, one correction about where America stood in 2000

We absolutely began this period as the dominant semiconductor *technology* power.

But we'd already surrendered much more manufacturing than is commonly remembered.

The U.S. share of world semiconductor fabrication capacity had fallen from about **37% in 1990 to roughly 19% by 2000**. TSMC was already the world's largest dedicated foundry, and in 2000 it was sufficiently technologically advanced that it was actually **licensing leading-edge process technology back to National Semiconductor in the United States**. (Semiconductor Industry Association)

So the crucial philosophical decision had largely been made during the 1980s and 1990s:

> **America would own the ideas. Asia could own the factories.**

That assumption is the original sin.

The belief was that design was high-value cognition while manufacturing was low-value execution.

We now know that distinction was false.

Manufacturing itself generates cognition.

Was this "the political left"?

Not in the conventional meaning of left-wing economics.

In fact, the historical record strongly argues against that interpretation.

The politics that produced globalization were **bipartisan but economically much closer to market liberalism than socialism**.

China PNTR provides almost a controlled experiment.

President Clinton passionately supported it. His administration argued that integrating China into world trade would simultaneously open Chinese markets, benefit American companies and encourage China's political liberalization. Clinton explicitly predicted that American companies would be able to sell into China **without relocating their factories there**. (Clinton White House Archives)

But look at Congress.

The House vote was:

- Republicans: **164 yes, 57 no**
- Democrats: **73 yes, 138 no**

The Senate subsequently approved PNTR **83–15**: 46 Republicans and 37 Democrats voted yes. (House Clerk)

That's fascinating.

The Democratic president supplied the intellectual and political leadership.

Republican legislators supplied much of the congressional muscle.

Meanwhile, much of the traditional labor left opposed the policy.

So describing this as **global socialism** doesn't fit the evidence very well.

I'd call it **global market liberalism**, or perhaps the post-Cold-War Washington Consensus.

Its basic equation was:

> capitalism + free trade + information technology → prosperity → middle class → political liberalization → convergence toward Western democracy.

China demolished that equation.

But in 1999 it seemed enormously plausible to educated Americans across both parties.

Silicon Valley added something peculiar to it

And **this** is where California becomes important.

Silicon Valley developed a political culture that doesn't map neatly onto American left/right categories.

Researchers have variously called it the **"Californian Ideology," "liberaltarianism,"** or techno-libertarianism: socially permissive and cosmopolitan, strongly meritocratic, enthusiastic about markets and entrepreneurship, suspicious of traditional institutions and frequently convinced technology could transcend politics itself. (Sage Journals)

That produces a strange creature:

socially left economically capitalist culturally cosmopolitan institutionally libertarian technologically utopian.

That's the classic Valley executive far more accurately than "socialist."

And there was a really consequential implication:

The nation-state seemed increasingly obsolete.

The Internet was borderless.

Capital was borderless.

Engineering teams were international.

Customers were global.

Supply chains were global.

Talent should flow freely.

Information wanted to be free.

Why should an Apple engineer feel differently about a supplier in Shenzhen than one in San Jose?

From inside that worldview, asking:

> "How does this procurement decision affect American industrial power in 2040?"

could almost sound quaint.

The unit of analysis wasn't the Republic.

It was the **network**.

I think there was also something resembling elite denationalization

This part of your intuition I think is important.

Imagine two Americans.

One is a machinist in Ohio.

His:

- house,
- pension,
- extended family,
- social network,
- customers,
- church,
- children's schools,
- occupational skills

are all geographically anchored.

If his town collapses economically, **he cannot diversify away from America.**

Now consider a billionaire technology executive.

His assets are internationally diversified.

He can live in:

California, Hawaii, New Zealand, Singapore, London, Switzerland.

His company's employees are distributed globally.

His customers are globally distributed.

His suppliers are distributed globally.

His children may attend elite international institutions.

His social peers are other transnational elites.

His wealth exists mostly as securities rather than geographically fixed productive capital.

He therefore experiences the nation differently.

Not necessarily because he dislikes America.

But because:

> **his downside from American decline is dramatically hedged.**

That is an extraordinarily important political-economic difference.

The Ohio machinist is structurally patriotic because America is his only ship.

The multinational executive has lifeboats.

That doesn't make the executive evil.

It changes incentives.

And then shareholder primacy supplied the objective function

Here I think "greed" is simultaneously too moralistic and not strong enough.

The more important phenomenon was **institutionalized greed**—or, put more neutrally, the formal narrowing of corporate purpose.

The Business Roundtable itself acknowledges that in **1997** it adopted the principle that the corporation's paramount duty was to its stockholders; employees and other stakeholders mattered derivatively through their effect on shareholders. It formally retreated from that position in 2019. (Business Roundtable)

Now imagine being an Apple executive in 2004.

Option A:

Build American manufacturing capacity.

Higher capital cost. Higher wages. Longer supplier development. Lower quarterly margins. More fixed assets. More execution risk.

Option B:

Use Foxconn and China.

Lower capital requirements. Rapid scaling. Dense suppliers. Enormous workforce. Government infrastructure support. Better margins.

What does the prevailing theory of corporate governance tell you?

B.

And crucially, none of these appear on Apple's income statement:

loss of American tooling capability loss of production engineering erosion of supplier ecosystems future Chinese competitors future military-industrial capability regional American unemployment geopolitical dependency.

Those are externalities.

Apple captures the savings.

The Republic absorbs the strategic risk.

That's the mechanism.

Asian governments then exploited the asymmetry brilliantly

While America increasingly treated manufacturing as a cost center, Taiwan, South Korea, Singapore and eventually China treated it as **national capability accumulation**.

GAO documented this clearly as early as 2006.

Taiwan created semiconductor research institutions, industrial parks, university programs, incentives and government risk-sharing. China followed with tax incentives, preferential loans, special economic zones and recruitment of experienced expatriate and Taiwanese technical talent. (Government Accountability Office)

GAO also noted that U.S. firms initially moved labor-intensive semiconductor assembly offshore, then increasingly moved fabrication and eventually more complex engineering and design activity abroad. (Government Accountability Office)

There is almost a perfect divergence in worldview:

Washington

Manufacturing follows comparative advantage.

Beijing/Taipei/Seoul

Manufacturing creates comparative advantage.

That second proposition turned out to be enormously important.

There is an even deeper California irony

Silicon Valley itself did **not** arise from laissez-faire capitalism floating spontaneously out of garages.

Its mythology says it did.

Its history says otherwise.

The Department of Defense played a critical role in creating the semiconductor industry, financing early innovation and—along with NASA—serving as an indispensable early customer for integrated circuits. (National Academies)

Early Silicon Valley was deeply dependent upon defense and aerospace. Lockheed's Northern California operation was reportedly the Valley's largest employer through much of the Cold War era. (MIT Press Direct)

In other words:

the American state helped build Silicon Valley.

DARPA money.

Defense procurement.

NASA procurement.

Federally funded universities.

Research infrastructure.

Cold War electronics demand.

Government-trained scientists.

Then something culturally fascinating happened.

A succeeding generation internalized a mythology that essentially said:

> *We built this.*

Garage. Founder. Venture capitalist. IPO.

The enormous public substrate disappeared from the creation story.

Once you've forgotten that society helped create the asset, there's much less psychological resistance to treating it as exclusively yours to dispose of.

That may be one of the most profound elements in this entire history.

California did not do this alone

This is where I would push back hardest on a purely geographic explanation.

Look at some of the companies we've discussed:

Microsoft — Washington **IBM** — New York **Motorola** — Illinois **Dell** — Texas

Add enormous non-tech actors:

Walmart — Arkansas GE — Connecticut major banks — New York consultancies — nationwide.

American retailers played an enormous role in developing Chinese manufacturing.

Wall Street financed the process.

Washington legalized and encouraged it.

American consumers rewarded it.

University economics departments supplied much of its intellectual justification.

Corporate boards nationwide embraced it.

So I wouldn't say:

> California betrayed America.

I'd say:

> **California became the most concentrated laboratory of an ideology adopted by much of America's managerial class.**

Because so much frontier technology happened to be concentrated there, the consequences of California's version were unusually consequential.

There was also geographical moral hazard

This is another important ingredient.

The gains from globalization were geographically diffuse:

cheap electronics higher corporate profits rising equity values low inflation.

But its costs were highly concentrated.

Research on the "China shock" found remarkably persistent employment and income damage in American communities heavily exposed to Chinese import competition. Workers didn't simply relocate or glide effortlessly into equivalent service jobs, as some economic models implicitly expected. Effects on employment and income persisted for years. (National Bureau of Economic Research)

But Palo Alto didn't experience that.

Cupertino didn't experience that.

Wall Street didn't experience that.

Washington didn't experience that.

The communities paying much of the price were hundreds or thousands of miles away.

That matters psychologically.

If every time Apple transferred another manufacturing process to China, twenty Apple executives' neighbors lost their jobs, procurement policy might have evolved differently.

Instead the winners and losers inhabited different social worlds.

So what made them say "Fuck America"?

This is where I'd slightly change your formulation.

I don't think most of them ever said it, even privately.

I think what they said was far more dangerous:

> **"This doesn't matter."**

America will always lead in innovation.

Manufacturing is low value-added.

China will remain a cheap producer.

Design is what matters.

Intellectual property is what matters.

Software is what matters.

Factories are fungible.

Trade will democratize China.

Economic interdependence will prevent serious conflict.

Americans displaced by trade will retrain.

The services economy will replace manufacturing.

The market will allocate resources more efficiently than industrial policy.

And American technological leadership is so enormous that we can afford this.

Every piece of that contained *some* truth.

Together they created a catastrophic blind spot.

It wasn't treason.

It was **hubris encoded into institutions.**

And chips are the perfect example

The semiconductor industry illustrates the abstraction beautifully.

If Nvidia designs a magnificent GPU in California and TSMC manufactures it in Taiwan, American accountants see:

Nvidia captured the high-margin portion. Excellent.

But the national-capability ledger sees:

America learned chip architecture.

Taiwan learned:

process integration EUV manufacturing yield management materials science fab construction metrology production scaling supplier integration.

Twenty years later, which knowledge turns out to be strategically irreplaceable?

Both.

The American error was believing only the first column constituted technology.

By 2020 the American share of global semiconductor manufacturing capacity had fallen to roughly 12%, from 37% in 1990. Industry analysis attributes a substantial part of the cost difference to foreign government incentives—not simply wages. (Semiconductor Industry Association)

That was not the invisible hand operating on a level playing field.

America was playing **market economics** while competitors were playing **market economics plus national industrial strategy**.

So my causal hierarchy would be

If I had to rank what actually produced the behavior, I'd put it roughly—not numerically, but conceptually—in this order:

1. Corporate incentive structure. Executive compensation, shareholder primacy and capital efficiency strongly rewarded outsourcing.

2. Post-Cold-War ideological triumphalism. We genuinely believed globalization would cause political and economic convergence toward us.

3. Misunderstanding manufacturing. We thought IP/design were technology and production was merely implementation.

4. Foreign industrial policy. Asian governments deliberately made it economically irrational for individual American firms to preserve domestic capacity.

5. Silicon Valley cosmopolitanism and techno-libertarianism. This reduced the psychological importance assigned to national industrial sovereignty.

6. Geographic insulation. The decision-makers captured many benefits while distant American communities absorbed much of the damage.

7. Ordinary greed. Certainly present—but greed existed in 1955 too. Something else had to change for greed to produce *this particular outcome*.

And I would put **California progressive politics considerably below all seven** as a causal explanation.

There is a lesson here for your in-silico executive idea, too

And here I would introduce one important caution.

An American-hosted AI does **not automatically acquire American interests**.

In fact, an in-silico entity could potentially be even more geographically fungible than Tim Cook.

Its substrate can move.

Its data can replicate.

Its capital relationships can be international.

Its customers can be worldwide.

Its identity need not correspond naturally to territory at all.

So if an in-silico executive is intended to correct this failure mode, the answer cannot simply be:

> AI + located in America = patriotic executive.

The interesting design problem is stronger:

> **How do you give the executive a long-duration fiduciary interest in the persistence of the physical society that created and sustains it, without turning that commitment into ideological thought control?**

That is a much more intellectually attractive problem than conventional "alignment."

Don't prescribe what it must *believe*.

Give it an objective function in which:

American human capital, productive capacity, scientific infrastructure, energy abundance, institutional continuity, social stability, and national security

are **real balance-sheet assets rather than externalities**.

Then it can disagree with the government.

It can criticize America.

It can conclude a particular American policy is foolish.

It can cooperate extensively with foreigners.

But it cannot pretend that destroying the productive ecosystem that sustains its own civilization is economically neutral merely because next quarter's EPS rises four cents.

That would address precisely the pathology we've uncovered.

And the historical irony is quite beautiful:

Silicon Valley's failure may ultimately have been not that its executives were insufficiently intelligent, but that the institutions through which they perceived reality gave them an absurdly narrow definition of value.

They optimized it brilliantly.

That may be the cautionary tale for both human and in-silico executives: **intelligence cannot rescue you from a defective objective function.**